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Globalizing Indian Thought

Week 27: Capstone-3 Assignment

Generative AI Applications in Business Functions

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Introduction

The following outlines the application of Generative AI across key business functions, specifically Marketing, Supply Chain, and Finance. It explores how Generative AI tools can be leveraged to enhance productivity, decision-making, and communication, while also addressing associated ethical risks. The analysis considers the potential of these tools to transform traditional processes and create new opportunities for innovation and efficiency across various industries.

Q1. Crafting Effective Prompts for Marketing

GenAI Prompt:

Compose a follow-up email to customers who abandoned their shopping carts on our e-commerce platform. The tone should be friendly and encouraging, with a hint of urgency. The audience is residential and SME users interested in energy efficiency and cost savings. The email should:

- Remind the user of the specific items left in their cart (e.g., "smart thermostat," "energy monitoring dashboard").
- Offer a 10% discount on those items, valid for the next 48 hours, to incentivize immediate completion of the purchase.
- Highlight the key energy-saving benefits of our smart energy management system (SEMS), such as reducing energy consumption by up to 20%, lowering electricity bills, and providing real-time insights into energy usage.
- Include a clear and compelling call to action to return to their cart and complete the purchase (e.g., "Click here to complete your purchase and start saving energy!").
- Specify the output format as a single paragraph.

Explanation:

Specific prompts work better than vague ones because they provide the GenAI model with clear constraints and context, leading to more relevant and targeted outputs. In this case, specifying the tone, audience, desired content, and output format ensures the generated email aligns with the marketing objectives. For example, specifying "friendly and encouraging, with a hint of urgency" will generate a different email than simply saying "friendly." Adding "valid for the next 48 hours" creates a sense of urgency. By including example items and quantifying the energy savings, the prompt becomes even more precise, leading to a more effective output.

- **Objective:** The primary objective is to increase e-commerce sales by converting abandoned shopping carts into completed purchases. This is achieved by using GenAI to create targeted and persuasive follow-up emails.
- **Proposed GenAI Use Case:** The proposed use case is to automate the creation of personalized follow-up emails for customers who abandon their shopping carts. This automation aims to improve customer engagement and drive conversions.
- **Tools & Technologies:** The core tool is a Generative AI model (like GPT-3.5) capable of natural language generation. The technology involves using this model to process the provided prompt and generate a marketing email.
- **Real-world Use Cases & Parallels:** Many e-commerce companies use automated email sequences to recover abandoned carts. Tools like Klaviyo, Mailchimp, and HubSpot offer features to create these emails, often with some level of personalization (though not always GenAI-powered). Parallels include personalized product recommendations, targeted advertising, and dynamic content generation, all of which use AI to tailor marketing messages.
- **Expected Business Impact:** The expected business impact includes increased conversion rates, higher revenue, improved customer retention, and reduced marketing costs through automation. By sending timely and relevant follow-up emails, businesses can recapture lost sales and improve the overall customer experience.

Q2. Decision Support with GenAI in Supply Chain

As a supply chain analyst, I would use GenAI in the following ways:

- ◆ **Extract and compare pricing, warranty, and support details:** I would use GenAI to parse the vendor proposals (which might be in different formats like PDF, Word, Excel, etc.) and extract the relevant information. A detailed prompt would specify the key data points: "Extract the following information from each vendor proposal: vendor name, product names, pricing per unit, total price, warranty duration, warranty terms and conditions (including coverage details), support availability (24/7, business hours, etc.), support response time (in hours), and any service level agreements (SLAs) related to support." The output could be structured as a table or JSON object with clear headings for each data point, facilitating easy comparison.
- ◆ **Generate a recommendation on which vendor to select:** I would provide GenAI with the extracted data and the company's priorities (e.g., lowest cost, longest warranty, best support, vendor reputation, component quality). A prompt like, "Based on the following data [provide extracted data] and the following priorities [list priorities, e.g., "Prioritize lowest total cost, followed by longest warranty duration. Support response time is important, but less critical than cost and warranty"], recommend a vendor and explain your reasoning. Also, provide a weighted score for each vendor based on these priorities."
- ◆ **Summarize the pros and cons of each proposal:** I would use GenAI to analyze the proposals and generate a concise summary of the advantages and disadvantages of each

vendor. A prompt such as, "Summarize the pros and cons of each vendor proposal, considering factors such as cost (including any hidden costs), warranty (coverage and exclusions), support (availability and responsiveness), vendor reputation, component quality, delivery timeline, and any other relevant factors. Format the output as a bulleted list for each vendor."

- **Objective:** The objective is to streamline and improve the vendor selection process in supply chain management, making it more efficient, objective, and data-driven.
- **Proposed GenAI Use Case:** The proposed use case is to use GenAI to automate the extraction of key information from vendor proposals, compare them based on specified criteria, and generate recommendations.
- **Tools & Technologies:** This involves using a combination of technologies:
 - GenAI for natural language processing (NLP) to extract information from documents.
 - Data structuring tools to organize the extracted data (e.g., JSON, tables).
 - Potentially, rule-based systems or other AI algorithms to apply the company's priorities and generate a weighted score.
- **Real-world Use Cases & Parallels:**
 - Many companies use contract management software with some AI capabilities to extract clauses and key data from legal documents.
 - Procurement platforms are starting to incorporate AI to analyze supplier data and provide insights.
 - Parallels include using AI in finance to analyze loan applications or in HR to screen resumes.
- **Expected Business Impact:**
 - Reduced time and effort in vendor selection.
 - Improved accuracy and objectivity in decision-making.
 - Lower procurement costs.
 - Reduced risk of choosing unsuitable vendors.
 - Better compliance with procurement policies.

Q3. Ethical Risk in Financial Automation

- ◆ **Benefit of using GenAI:** GenAI can significantly speed up the process of generating compliance reports, reducing the time and resources required for manual report preparation. This allows finance teams to focus on higher-level analysis and decision-making, such as identifying trends, investigating anomalies, and developing strategies to improve financial health. It can also reduce the risk of human error in report preparation.
 - ◆ **Ethical risk or limitation:** One ethical risk is the potential for bias in the data used to train the GenAI model. If the training data reflects existing biases in financial reporting (e.g., underreporting of certain types of expenses, misclassification of revenue), the GenAI system may perpetuate or even amplify these biases in its generated reports, leading to unfair or inaccurate assessments of compliance. Another risk is the lack of transparency in how the GenAI model arrives at its conclusions, making it difficult to identify and correct errors or biases. Over-reliance on GenAI without human oversight can also lead to a lack of accountability.
 - ◆ **Safeguard or best practice:** A critical safeguard is to ensure the transparency and auditability of the GenAI system. This includes:
 - Maintaining detailed records of the data used to train the model, including its source, quality, and any preprocessing steps.
 - Documenting the algorithms employed, including their limitations and assumptions.
 - Implementing a robust validation process, including human review and comparison with traditional reporting methods.
 - Establishing clear lines of responsibility and accountability for the reports generated by the GenAI system.
 - Regularly auditing the GenAI system for bias and accuracy.
 - Developing a comprehensive ethical framework for the use of GenAI in finance, including guidelines for data privacy, security, and fairness.
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- **Objective:** The objective is to automate the generation of compliance reports in finance, increasing efficiency and reducing manual effort.
 - **Proposed GenAI Use Case:** The proposed use case is to employ GenAI to automatically compile and format data from various sources into regulatory compliance reports.
 - **Tools & Technologies:**
 - GenAI for NLP to understand and process financial documents and regulations.
 - Data integration tools to gather data from different financial systems.
 - Reporting software to format the final reports.

- **Real-world Use Cases & Parallels:**

- Some financial institutions use AI-powered tools for fraud detection and anti-money laundering (AML).
- Tax preparation software uses AI to help individuals and businesses file their taxes accurately.
- Parallels exist in legal tech, where AI is used for e-discovery and contract analysis.

- **Expected Business Impact:**

- Significantly reduced time and cost for compliance reporting.
- Improved accuracy and consistency of reports.
- Freeing up finance staff for more strategic tasks.
- Potential for real-time compliance monitoring.